Boiler Plate Thickness Calculation

Problem

M2:A cylindrical boiler operates at a pressure of 12 bar and has an internal diameter of 3 m. The maximum allowable tensile stress in the shell plate is 90 MN/m^2. Determine the minimum required plate thickness for the following cases:

A. When the joints are riveted with a joint efficiency of 70%

The formula for calculating the plate thickness is:

$$t = \frac{P \cdot d}{2 \cdot \sigma \cdot \eta}$$

Where:

- P = internal pressure $= 12\,\text{bar} = 12 \times 10^5\,\text{N/m}^2$
- d = internal diameter $= 3\,\text{m}$
- σ = maximum allowable tensile stress $= 90 \times 10^6\,\text{N/m}^2$
- η = joint efficiency $= 0.7$

Substituting the values:

$$t = \frac{12 \times 10^5 \cdot 3}{2 \cdot 90 \times 10^6 \cdot 0.7}$$

Simplify:

$$t = \frac{3.6 \times 10^6}{126 \times 10^6}$$

$$t = 0.02857\,\text{m} = 28.57\,\text{mm}$$

Thus, the minimum required plate thickness is **28.57 mm**. **(3 marks)**.

B. When the joints are welded

For welded joints, the joint efficiency is $\eta = 1$. Using the same formula:

$$t = \frac{P \cdot d}{2 \cdot \sigma \cdot \eta}$$

Substituting the values:

$$t = \frac{12 \times 10^5 \cdot 3}{2 \cdot 90 \times 10^6 \cdot 1}$$

Simplify:

$$t = \frac{3.6 \times 10^6}{180 \times 10^6}$$

$$t = 0.02\,\text{m} = 20\,\text{mm}$$

Thus, the minimum required plate thickness is **20 mm**. **(2 marks)**.